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TEST REPORT

Lucideon Reference: N26458 (QT-78966/4/KNA)/Ref. 2

Project Title: Flexural Testing to BS EN 1052-2:2016 of N10 Carbon Negative Bricks

Client: Earth4Earth Technology Ltd
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Report Date: 26 May, 2026

Purchase Order No.: PO-0003

Opinions and interpretations expressed herein are outside the scope of UKAS Accreditation.

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AMRICC INSIDE

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CONTENTS

1 INTRODUCTION

Earth4Earth Technology Ltd required a programme of testing to determine the flexural strength of masonry samples comprising of N10 Carbon Negative bricks in accordance with BS EN 1052-2:2016 Methods of Test for Masonry – Part 2: Determination of Flexural Strength.

Testing was carried out on 1 May 2026 at Lucideon's Structures Laboratory at Queens Road, Penkhull, Stoke-on-Trent, ST4 7LQ.

2 SAMPLES RECEIVED

Solid bricks referenced as N10 Carbon Negative were received from the client.

The bricks were nominally 215 mm x 102.5 mm x 65 mm (length x width x height).

The bricks were solid with a nominal compressive strength of 39.1 N/mm².

3 METHOD OF CONSTRUCTION

The specimens to be tested in parallel orientation were built 10 courses high by 2 courses wide and specimens tested in perpendicular orientation were built 4 courses high by 4 courses wide using M4 mortar.

The specimens were close covered with polythene sheets for 28 days at ambient laboratory temperatures with a pre-compression of 2.70 x 10^{-3} N/mm^2.

The specimens were constructed on 2 April 2026.

4 METHOD OF TEST

All specimens were tested in two purpose made 4-point loading test rigs: one rig to test specimens loaded parallel to the specimen bed joint and one rig to test specimens loaded perpendicular to the specimen bed joint.

For the tests carried out parallel to the bed joint, the inner bearers were set at a distance of 53% of the outer bearers i.e. the inner bearers were 355 mm, and the outer bearers were 670 mm.

For the tests perpendicular to the bed joint, the inner bearers were set at 41% of the outer bearers i.e. the inner bearers were set at 330 mm and the outer bearers were set at 800 mm.

It was ensured that the base of each specimen tested was in contact with two non-friction layers of material (Teflon™).

Loading was carried out at a loading rate of approximately 0.1 N/mm² until failure.

The load was recorded using a load cell calibrated to National Standards and the data was collected using a calibrated Data Logger and laptop.

All specimens were tested in a vertical direction.

Testing was carried out at 19°C and 48% relative humidity.

5 RESULTS

Results for all testing carried out can be seen in the Tables Section.

6 SUMMARY

For testing carried out in the parallel orientation, N10 Carbon Negatives achieved a characteristic value of 0.03 N/mm^2 in accordance with BS EN 1052-2:2016.

For testing carried out in perpendicular orientation, N10 Carbon Negatives achieved a characteristic value of 0.26 N/mm^2 in accordance with BS EN 1052-2:2016.

TABLES

Table 1 – Mortar Properties

*Compressive Strength to EN 1015-11 (N/mm^2) *	*Air Content to EN 1015-7 (%)	*Flow Value to BS 4551
4.32	5.2	173

**These tests are not on Lucideon's UKAS Schedule.*

Table 2 – Results for N10 Carbon Negatives Tested in Parallel Orientation

Sample Number	Length (mm)	Breadth (mm)	Depth (mm)	Failure Load (N)	Flexural Strength at Failure (N/mm^2)	Comments
1	745	441	105	230	0.02	Flexural failure
2	746	440	105	490	0.05	Flexural failure
3	745	442	105	750	0.07	Flexural failure
4	746	441	103	460	0.05	Flexural failure
5	747	440	104	500	0.05	Flexural failure
Mean	**745.8**	**440.8**	**104.4**	**486**	**0.05**	**-**

Table 3 – Results for N10 Carbon Negatives Tested in Perpendicular Orientation

Sample Number	Length (mm)	Breadth (mm)	Depth (mm)	Failure Load (N)	Flexural Strength at Failure (N/mm^2)	Comments
1	900	300	105	2140	0.46	Flexural failure
2	900	301	104	1820	0.39	Flexural failure
3	901	300	105	1730	0.37	Flexural failure
4	902	302	105	1600	0.34	Flexural failure
5	901	300	104	1860	0.40	Flexural failure
Mean	**900.8**	**300.6**	**104.6**	**1830**	**0.39**	**-**

NOTE: The results given in this report apply only to the samples that have been tested.

END OF REPORT

PLATES

Plate 1 – Flexural Failure in Parallel Orientation

Plate 2 – Flexural Failure in Perpendicular Orientation